Product Data
Series 4020
Computerized Gas Mixing and Dilution System

A low cost, computerized gas blending and gas dilution system with an advanced Windows user interface.

The Environics® Series 4020 is a computerized gas mixing/dilution system which offers the unique capability of mixing or diluting gases in a single chassis. The gas mixes can be used in generating precise gas calibration standards, creating gaseous atmospheres or producing gas mixes for analytical research or production purposes. The Series 4020 can produce gas concentrations from percent to ppb levels for single or multi-point calibration.

The Series 4020 can mix up to seven (7) component gases simultaneously to generate precise gas calibration standards and gaseous atmospheres in a broad dynamic range. The dilution capabilities of the 4020 allows from 1:1 to 10,000:1 depending on the configuration of the system.

The system consists of two components: the Series 4020 instrument and the user's personal computer. The user interface is a Microsoft Windows application that communicates with the Environics system via USB or an optional RS232 interface.

The Series 4020 consists of a single chassis supporting up to eight mass flow controllers. The mass flow controllers are factory calibrated using a primary flow standard traceable to the United States National Institute of Standards and Technology (NIST). Each flow controller utilizes an 11 point calibration table with linear interpolation, to increase accuracy and reduce flow controller nonlinearity.

PRODUCT FEATURES AND BENEFITS

- Broad range of dilution ratios (up to 10,000:1) allows the user to significantly reduce the number of cylinders needed to perform compliance tests.
- Allows multi-point calibration of analyzers.
- Automatic calculation of dilution and span gas flows.
- Internally-stored Mass Flow Controller calibration tables .
- Modular design allows for additional gas circuits to be added.
- User definable cylinder library allows for easy selection of frequently used gas cylinders. Each cylinder may contain an unlimited number of component gases with automatic K-factor calculation.

INSTRUMENT OPERATION

The Series 4020 has four basic modes of operation.

- Concentration Mode: Allows user to create a blend by entering target gas concentrations for each source gas cylinder, and the desired total output flow for the mix.
- Divider Mode: Allows the user to operate the instrument as an automated ten step gas divider.
- Flow Mode: Allows user to specify the flow rate of each gas cylinder.
- Program Mode: Provides the ability to program the instrument for unattended operation. Programs can be recalled and run in any sequence, at various times/dates.

SPECIFICATIONS

Performance

Accuracy
Concentration: ± 1.0% setpoint
Flow: ± 1.0% setpoint
Repeatability: ± 0.05% setpoint

*Performance specifications are valid when all Mass Flow Controllers are operating between 10% and 100% of full scale flow. Mass flow controllers are calibrated using a NIST traceable Primary Flow Standard, using a Reference Temperature of 0° C (32°F) and a Reference Pressure of 760 mm Hg (29.92 in. Hg)

Warm up time: 30 minutes

Mechanical

Inlets
External 1/4" Swagelok (or compatible fitting)

Outlet
External 1/4" Swagelok (or compatible fitting)

Operating Pressures at inlets
Recommended: 25 psig (1.68 Bar)
Minimum: 10 psig (0.67 Bar)
Maximum: 75 psig (5.04 Bar)

Wetted Surfaces
Tubing: Electropolished 316 Stainless Steel
MFC's: Stainless Steel
Seals: Viton
(Optional: Kalrez, Buna-N, Neoprene, Metal)

Operating temperatures
32°- 104° F (0° - 40° C)

Performance temperatures
59° - 95° F (15° - 35° C)

Weight
18 lbs. (2 Mass Flow Controllers)
35 lbs. (8 Mass Flow Controllers)

Dimensions (w x h x d)
Portable: 17" x 7" x 15"
(43.18cm x 17.78 cm x 38.1 cm)

Rack: 19" x 7" x 15"
(48.26cm x 17.78 cm x 38.1 cm)
Portable: 17" x 7" x 15"
(43.18cm x 17.78 cm x 38.1 cm)

Electrical

110 to 240 VAC, 50/60 Hz

Electronics

12 Bit A/D and D/A Conversion
USB (or RS232 Serial interface, optional)

Software

Environics Instrument Control Software
(supplied on CD-ROM or downloaded)
PC Requirements
IBM PC or compatible (486-33 or higher)
Microsoft Windows 05/98/Me/NT/2000/XP or Windows 7/8/10
8 MB RAM
10 MB Hard Disk Space
USB (or RS-232 Communication port, optional)

OPTIONS

Additional MFC's (up to 8)

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